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To cite this article: Allanaboyina V. V. S. Durga Prasad & Amit K. Dhiman (2014) CFD Analysis of Momentum and Heat Transfer Around a Pair of Square Cylinders in Side-by-Side Arrangement, Heat Transfer Engineering, 35:4, 398-411, DOI: [10.1080/01457632.2013.828561](https://doi.org/10.1080/01457632.2013.828561)

To link to this article: <https://doi.org/10.1080/01457632.2013.828561>



Published online: 30 Sep 2013.



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# CFD Analysis of Momentum and Heat Transfer Around a Pair of Square Cylinders in Side-by-Side Arrangement

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*A numerical investigation is conducted to analyze the steady and unsteady laminar flow and heat transfer in a channel with two square bars (of equal diameters) arranged side-by-side. The analysis is carried out for Reynolds number = 10–100, Prandtl number = 0.7–50, for transverse separation distance between the bars (i.e., gap ratio) of 1.5, 2, 2.5, 5, and 10 at the blockage ratio of 1/18. The results found here are in good agreement with previously published data. The effects of gap ratio, Reynolds number, and Prandtl number on the detailed kinematics of the flow and the heat transfer are presented. The engineering parameters such as total drag coefficient, average Nusselt number, and Strouhal number are calculated for the preceding range of conditions. It is observed that the overall drag coefficient decreases with increasing Reynolds number, whereas the average Nusselt number for the square cylinders increases with increasing Reynolds number and/or Prandtl number for all the values of gap ratios studied. The percentage enhancement in the value of the average Nusselt number is found to be more than 76% for the range of settings covered here.*

## INTRODUCTION

Flow around an array of bluff bodies is of practical engineering importance and has attracted a considerable amount of interest in recent years. The present analysis is relevant to many practical engineering applications. For instance, cylinders with a square/noncircular cross section are generally encountered in compact heat exchange systems, cooling towers, chimney stacks, wind turbine farms, electrical poles, solar heaters, electronic cooling, oil and gas pipelines, flow metering devices, and so forth. In particular, a submerged structure in the near-shore region is composed of an array of cylinders rather than an isolated cylinder. In contrast to the overwhelming number of investigations on the flow past circular cylinders, the square counterpart has not been investigated in the same extent, although it plays a dominant role in many engineering applications. The present study is concerned with the momentum and heat transfer around

two identical square cylinders in side-by-side arrangement in both steady and unsteady regimes. The relevant literature is discussed next for the problem under consideration.

Valencia and Paredes [1] conducted a two-dimensional (2-D) numerical investigation to analyze the unsteady laminar flow and heat transfer characteristics in a plane channel with two square bars mounted side-by-side to the approaching flow for a transverse separation ratio or gap ratio ( $s/d$ ) ranging from 0 to 5 for Reynolds number ranging from 25 to 125 at a fixed blockage ratio of 1/8. The numerical results reveal the complex structure of the flow. The flow is steady at  $Re = 200$ ; it becomes (periodic) unsteady as the Reynolds number is further increased, and shows low frequency modulation for  $Re > 400$ ; and the flow has a transitional character at  $Re = 1000$ . Peng [2] studied the flow behind two side-by-side square solids with a fixed gap ratio of 2 in the unconfined domain by both numerical simulation and experimental flow-visualization methods at a constant Reynolds number of 100. His findings show the bistability flows, with both in-phase and anti-phase synchronizations, by modifying the initial conditions. Agrawal et al. [3] investigated the flow around a pair of side-by-side square cylinders using the lattice Boltzmann method, revealed the existence of regimes

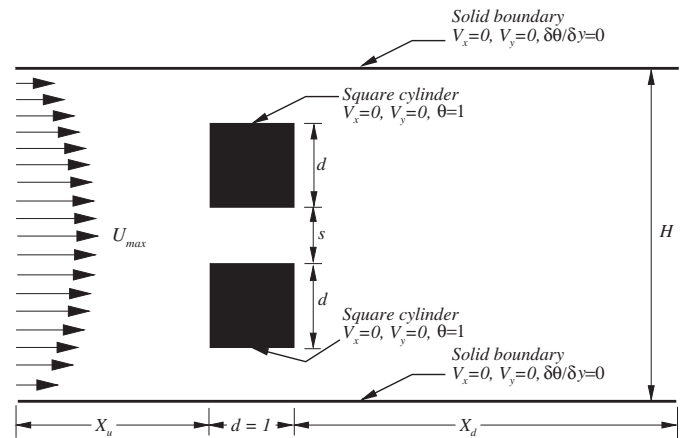
Address correspondence to Dr. Amit Kumar Dhiman, Department of Chemical Engineering, Indian Institute of Technology Roorkee, Roorkee–247 667, India. E-mail: dhimuamit@rediffmail.com

with either synchronized or nonsynchronized vortex shedding, and discussed the type of vortex shedding from the cylinder in both regimes for a uniform inlet velocity field. They reported their numerical results for two gap ratios ( $s/d$ ) of 0.7 and 2.5 for the fixed Reynolds number of 73 and the fixed blockage ratio of 1/18. Subsequently, Rao et al. [4] extended the work of Agrawal et al. [3] and conducted numerical analysis of the flow around a pair of side-by-side square cylinders using the lattice Boltzmann method, for separation ratios ranging from 1 to 2.7 for the Reynolds numbers ranging from 73 to 200 at the blockage ratio of 1/18. Their investigation reveals that for  $s/d < 1.5$ , the flow shows a flip-flop regime; however, for  $s/d > 1.5$ , the flow becomes synchronized. The transition between flip-flop and synchronized regimes occurs at  $s/d = 1.5$ . It is worthwhile to mention here that no information on heat transfer characteristics around side-by-side square cylinders is given by Agrawal et al. [3] and Rao et al. [4].

Moreover, sufficient information is available in the literature on the flow around a pair of side-by-side square cylinders in the turbulent regime. For instance, Wong et al. [5], Kolar et al. [6], Yen and Liu [7], and Alam et al. [8] experimentally investigated the wake flow around a single square cylinder and two side-by-side square cylinders. Sugun et al. [9], and Kumar and Vengadesan [10] conducted a numerical investigation to simulate the flow around two square cylinders arranged side-by-side.

For the row of square cylinders arranged in side-by-side arrangement, Mizushima and Akinaga [11] investigated interactions of wakes in a flow past a row of square bars by both numerical simulations and experiments on the assumption that the flow is 2-D and incompressible. Kumar et al. [12] reported the presence of synchronous, quasi-periodic, and chaotic flow regimes for  $0.3 \leq s/d \leq 12$  for nine square cylinders in side-by-side configuration at a fixed Reynolds number of 80. Along the same line, Sewatkar et al. [13] studied the combined effects of cylinder spacing and Reynolds number on the flow across a row of cylinders for  $30 \leq Re \leq 140$  and  $1.0 \leq s/d \leq 4.0$ . Chatterjee et al. [14] carried out numerical simulation of the flow around a row of five square cylinders placed in a side-by-side arrangement for gap ratios of 1.2, 2, 3, and 4 and normal to the oncoming flow at a fixed Reynolds number of 150.

Thus, from the preceding discussion, it can be summarized that virtually no work is available on the confined flow around a pair of two side-by-side bars of square cross section at low Reynolds numbers. Likewise, no heat transfer study is carried out across a pair of side-by-side square bluff bodies. As the multiple bluff obstacles create a complex flow and thermal structures even at low Reynolds numbers, and owing to the engineering relevance in various applications (mentioned in the first paragraph of this section), the understanding of flow and thermal patterns is essential. Hence, the aim here is to investigate the effects of Reynolds number ( $Re$ ), separation or gap ratio ( $s/d$ ), and Prandtl number ( $Pr$ ) on both flow and heat transfer over a pair of side-by-side square cylinders for  $s/d = 1.5-10$ ,  $10 \leq Re \leq 100$ , and  $Pr = 0.7-50$ .



**Figure 1** Schematic representation of the flow and heat transfer over a pair of square cylinders in side-by-side arrangement.

## PROBLEM STATEMENT AND MATHEMATICAL FORMULATION

In this study, the 2-D confined flow of an incompressible fluid in a channel with a pair of cylinders of square cross-section arranged side-by-side is studied, as shown in Figure 1. The obstacles (i.e., two square cylinders) are identical in shape and size ( $d$ ). These two obstacles are exposed to a fully developed velocity field with a maximum velocity ( $U_{max}$ ) and uniform temperature ( $T_{\infty}$ ) at the channel inlet. The two square bars are maintained at a constant temperature of  $T_w (> T_{\infty})$ . These bars are located at an upstream distance of  $X_u/d = 8.5$  from the inlet and at a downstream distance of  $X_d/d = 16.5$  from the outlet (Dhiman et al. [15, 16]). The total length of the computational domain is  $L/d = 26$  in the axial direction; however, the height of the domain is  $H/d = 18$  in the lateral direction. The value of the computational domain height is chosen based on the literature by Agrawal et al. [3] and Rao et al. [4]. The blockage ratio here is defined as  $\beta = d/H$ .

The flow and temperature fields are governed by continuity, Navier–Stokes, and energy equations. The fluid properties are assumed to be constant and the viscous dissipation in the energy equation is also neglected. For instance, the temperature difference between the heated square bars and the incoming fluid, that is,  $(T_w - T_{\infty})$ , is assumed to be small ( $= 2$  K) with  $T_w = 300$  K and  $T_{\infty} = 298$  K so that the variation of physical properties (e.g., density, viscosity, thermal conductivity, etc.) with temperature can be neglected. Therefore, the present results are applicable to situations where the temperature difference is not too large and for moderate viscosity. For the 2-D incompressible flow, the nondimensional form of the continuity,  $x$ -component and  $y$ -component of momentum, and energy equations are given by Eqs. (1)–(4).

Continuity equation:

$$\frac{\partial V_x}{\partial x} + \frac{\partial V_y}{\partial y} = 0 \quad (1)$$

*x*-Momentum equation:

$$\frac{\partial V_x}{\partial t} + \frac{\partial(V_x V_x)}{\partial x} + \frac{\partial(V_y V_x)}{\partial y} = -\frac{\partial p}{\partial x} + \frac{1}{\text{Re}} \left( \frac{\partial^2 V_x}{\partial x^2} + \frac{\partial^2 V_x}{\partial y^2} \right)$$

*y*-Momentum equation:

$$\frac{\partial V_y}{\partial t} + \frac{\partial(V_x V_y)}{\partial x} + \frac{\partial(V_y V_y)}{\partial y} = -\frac{\partial p}{\partial y} + \frac{1}{\text{Re}} \left( \frac{\partial^2 V_y}{\partial x^2} + \frac{\partial^2 V_y}{\partial y^2} \right)$$

Energy equation:

$$\frac{\partial \theta}{\partial t} + \frac{\partial(V_x \theta)}{\partial x} + \frac{\partial(V_y \theta)}{\partial y} = \frac{1}{\text{Re Pr}} \left( \frac{\partial^2 \theta}{\partial x^2} + \frac{\partial^2 \theta}{\partial y^2} \right) \quad (4)$$

The dimensionless boundary conditions may be written as follows:

- *At the inlet boundary:* The fully developed velocity profile is used,

$$V_x = 1 - (|1 - 2\beta y|)^2 \quad (0 \leq y \leq H/d, \beta = d/H);$$

$$V_y = 0, \theta = 0,$$

- *On the surface of the square cylinders:* The standard no slip condition is used,

$$V_x = 0, V_y = 0, \theta = 1,$$

- *On upper and lower boundaries,*

$$V_x = 0, V_y = 0, \frac{\partial \theta}{\partial y} = 0,$$

- *At the exit boundary:* The default outflow boundary condition in a commercial code [17, 18], which assumes a zero diffusion flux for all flow variables, is used and is chosen based on the information available in the literature [14, 19]. This choice implies that the conditions of the outflow plane are extrapolated from within the domain and as such have negligible influence on the upstream flow conditions. This is identical to homogeneous Neumann boundary condition, that is,

$$\frac{\partial V_x}{\partial x} = 0, \frac{\partial V_y}{\partial x} = 0, \frac{\partial \theta}{\partial x} = 0$$

The important dimensionless control parameters (Re and Pr) and output parameters ( $C_{DP}$ ,  $C_{DF}$ ,  $C_D$ ,  $C_L$ , and St) used in the present work are as follows:

- Reynolds number (Re) and Prandtl number (Pr) are defined as

$$\text{Re} = \frac{dU_{\max} \rho}{\mu} \quad \text{and} \quad \text{Pr} = \frac{\mu c_p}{k}$$

- Total drag coefficient ( $C_D$ ), the sum of the friction and pressure components, is defined as

$$C_D = \frac{F_D}{(1/2)\rho U_{\max}^2 d} = C_{DP} + C_{DF}$$

where  $F_D$  is the total drag force per unit length of the cylinder.

- Individual drag coefficients ( $C_{DP}$  and  $C_{DF}$ ) are evaluated by using the following definitions:

$$C_{DP} = \frac{F_{DP}}{(1/2)\rho U_{\max}^2 d} = \int_S C_p n_x dS \quad \text{and}$$

$$C_{DF} = \frac{F_{DF}}{(1/2)\rho U_{\max}^2 d} = \frac{2}{\text{Re}} \int_S (\tau \cdot n_s) dS$$

where  $n_s$  is the unit normal vector on the surface of the cylinder and  $n_x$  is its *x*-component.

- The total lift coefficient ( $C_L$ ) is calculated as

$$C_L = \frac{F_L}{(1/2)\rho U_{\max}^2 d}$$

where  $F_L$  is the total lift force per unit length of the cylinder.

- The Strouhal number is defined as

$$\text{St} = \frac{fd}{U_{\max}}$$

where  $f$  is the frequency of vortex shedding.

## GRID STRUCTURE AND NUMERICAL METHODOLOGY

The computational grid is generated by using the commercial computational fluid dynamics (CFD) preprocessor GAMBIT; it has both uniform and nonuniform grid distributions, having a fine grid in the region of larger gradients and coarse grid in the region of low gradients [17]. The computational grids consist of 76,360, 79,280, 82,200, 85,120, and 88,040 cells for  $s/d = 1.5, 2, 2.5, 5$ , and  $10$ , respectively (with each side of the square cylinder having 100 grid points with a fine grid size ( $\delta$ ) of  $0.01d$  near the cylinders, and near top and bottom walls of the channel and the largest grid size of  $0.5d$  away from cylinders and walls) and are found adequate for the generation of results. For instance, the grid resolution study is carried out to investigate the effect of grid size on the output dimensionless parameters for  $\text{Re} = 100$ ,  $\text{Pr} = 50$ , and  $s/d = 10$ . Three nonuniform grids of grid sizes 36,000, 88,040 and 179,020 cells are used with  $\delta$  of  $0.1d$ ,  $0.01d$ , and  $0.008d$ , respectively, for the largest grid size of  $0.5d$ .

The percentage relative differences in the values of total drag coefficient and Strouhal number are found to be about 2.8% and 3.7% for the grid of 36,000 cells with respect to the finest grid of 179,020 cells, respectively. Similarly, the corresponding differences in the values of total drag coefficient and Strouhal number are found to be only about 0.5% and 0.8%, respectively, for the grid of 88,040 cells. Thus, the grid size of 88,040 cells is found adequate for the generation of new results at  $s/d = 10$ .

The 2-D, steady/unsteady, laminar, pressure-based Navier–Stokes (pbns) solver module of Fluent is used to solve the incompressible laminar flow on the collocated grid arrangement [17, 18]. ANSYS has acquired a number of companies since 2000, including ICFM CFD Engineering, CADOE, CFX (2003), Century Dynamics, Harvard Thermal, Fluent, Inc. (2006), Ansoft Corporation (2008), Apache Design Solutions (2011), and Esterel Technologies (2012) [20]. ANSYS offers a comprehensive range of engineering simulation solution sets providing access to virtually any field of engineering simulation that a design process requires. Companies in a wide variety of industries use ANSYS software. The tools put a virtual product through a rigorous testing procedure (such as crashing a car into a brick wall, or running for several years on a tarmac road) before it becomes a physical object. The semi-implicit method for the pressure-linked equations (SIMPLE) scheme is used for solving pressure–velocity decoupling. The constant density and Newtonian viscosity model are used. However, the fully developed velocity profile is incorporated at the channel inlet by using user-defined functions (UDFs) available in Fluent. The second-order upwind scheme is used to discretize the convective term, while the diffusive term is discretized by a central difference scheme. The second-order implicit time-integration method is used here and the dimensionless time step is set to 0.01, as the smaller value of the time step did not produce any significant change in the values of the output dimensionless parameters considered in this work. The resulting algebraic equations are solved by the Gauss–Siedel point-by-point iterative method in conjunction with the algebraic multi-grid (AMG) solver. The use of the AMG scheme can greatly reduce the number of iterations and thus the central processing unit (CPU) time required to obtain a converged solution. The residuals of continuity,  $x$ - and  $y$ -components, and energy equations used are of the order of  $10^{-15}$  in the steady regime and of the order of  $10^{-20}$  each in the time-periodic regime.

## RESULTS AND DISCUSSION

In the present study, momentum and heat transfer results are generated in the full computational domain for the following range of settings: gap ratio ( $s/d$ ) = 1.5, 2, 2.5, 5, and 10; Reynolds number ( $Re$ ) = 10–100 (in intervals of 10); and Prandtl number ( $Pr$ ) = 0.7 and 50, for the fixed blockage ratio of 1/18.

The flow here is characterized by six dimensionless groups, namely, drag coefficient, Nusselt number, lift coefficient,

Reynolds number, Prandtl number, and Strouhal number. In the (periodic) unsteady regime, numerical calculation is carried out for at least 10 cycles after the asymptotic shedding frequency of the Karman vortex has been reached for the calculation of time-averaged flow and heat transfer output parameters.

## Validation of Results

The validation of numerical results is carried out by comparing the present results for a single square cylinder against the published data in the literature. The numerical results of Agrawal et al. [3] and Breuer et al. [21] on drag coefficient, lift coefficient, and Strouhal number are used as benchmark in the present study. Breuer et al. [21] investigated the confined flow around a square bar mounted inside a plane channel with a fixed blockage ratio of 1/8 by using two numerical techniques, namely, a lattice-Boltzmann automata and a finite-volume method for the Reynolds number range  $0.5 \leq Re \leq 300$ . An excellent agreement between present and literature values is observed for a single square cylinder, as illustrated in Table 1. For instance, the percentage deviations in the values of total drag coefficients compared with that of Breuer et al. [21] are found to be about 0.93%, 3.26%, and 1.58%, corresponding to Reynolds numbers of 60, 73, and 100, respectively. Similarly, the deviations in values of difference between maximum and minimum lift coefficients (i.e.,  $\text{Max } C_L - \text{Min } C_L$ ) are found to be about 0.15%, 3.80%, and 1.26% corresponding to Reynolds numbers of 60, 73, and 100, respectively. The corresponding differences in the value of the Strouhal number are about 0.57%, 1.26%, and 0.86% for  $Re = 60, 73$ , and 100, respectively. However, the relative differences in the present values of Strouhal number compared with that of Agrawal et al. [3] are found to be about 1.98%, 1.09%, and 4.62%, corresponding to  $Re = 60, 73$ , and 100, respectively (Table 1). Further, in the steady regime, the present values of the recirculation length with the results reported by Breuer et al. [21] and Dhiman et al. [22] are compared for a single square cylinder for  $Re = 40$  and blockage ratio of 1/8. The present dimensionless value of the recirculation length is found to be about 2.18, which is in excellent agreement with

**Table 1** Comparison of present results of  $C_D$ ,  $\text{Max } C_L - \text{Min } C_L$ , and  $St$  with literature values at different values of  $Re$  for a single square cylinder

| Source             | $Re$ | $C_D$ | $\text{Max } C_L - \text{Min } C_L$ | $St$  |
|--------------------|------|-------|-------------------------------------|-------|
| Present work       | 60   | 1.514 | 0.065                               | 0.119 |
| Breuer et al. [21] |      | 1.50  | 0.07                                | 0.120 |
| Agarwal et al. [3] |      | —     | —                                   | 0.117 |
| Present work       | 73   | 1.446 | 0.241                               | 0.127 |
| Breuer et al. [21] |      | 1.40  | 0.25                                | 0.129 |
| Agarwal et al. [3] |      | 1.376 | —                                   | 0.126 |
| Present work       | 100  | 1.371 | 0.395                               | 0.138 |
| Breuer et al. [21] |      | 1.35  | 0.40                                | 0.139 |
| Agarwal et al. [3] |      | —     | —                                   | 0.145 |

**Table 2** Validation of present results with literature values at different values of Re for side-by-side square cylinders

| Source                   | Re | $C_D$        |                 | $C_L$        |                 |
|--------------------------|----|--------------|-----------------|--------------|-----------------|
|                          |    | Top cylinder | Bottom cylinder | Top cylinder | Bottom cylinder |
| Present work             | 25 | 5.839        | 5.839           | 2.454        | -2.454          |
| Valencia and Paredes [1] |    | 5.698        | 5.698           | 2.389        | -2.389          |
| Present work             | 50 | 4.645        | 4.689           | 1.6193       | -1.687          |
| Valencia and Paredes [1] |    | 4.653        | 4.677           | 1.6364       | -1.668          |

the values of wake lengths of 2.15 and 2.17 of Breuer et al. [21] and Dhiman et al. [22], respectively.

On the other hand, the validation of the present results for a pair of side-by-side square cylinders is carried out with the results of Valencia and Parades [1], for the fixed blockage ratio of 1/8 and at the constant gap ratio of 0.75 (Table 2). The results of total drag coefficient and total lift coefficient for both top and bottom cylinders show that excellent agreement exists between the two studies. The deviations of the total drag coefficients are found to be about 2.5% for both top and bottom cylinders, corresponding to Re = 25 (steady-state result); however, deviations are about 0.16% for a top cylinder and about 0.27% for a bottom cylinder, corresponding to Re = 50 (unsteady-state result). Similarly, the corresponding differences in the values of total lift coefficients are about 2.7% for both top and bottom cylinders corresponding to Re = 25, and about 1.04% for a top cylinder and about 1.15% for a bottom square cylinder corresponding to Re = 50, respectively. This validates the present numerical solution procedure.

### Flow Patterns

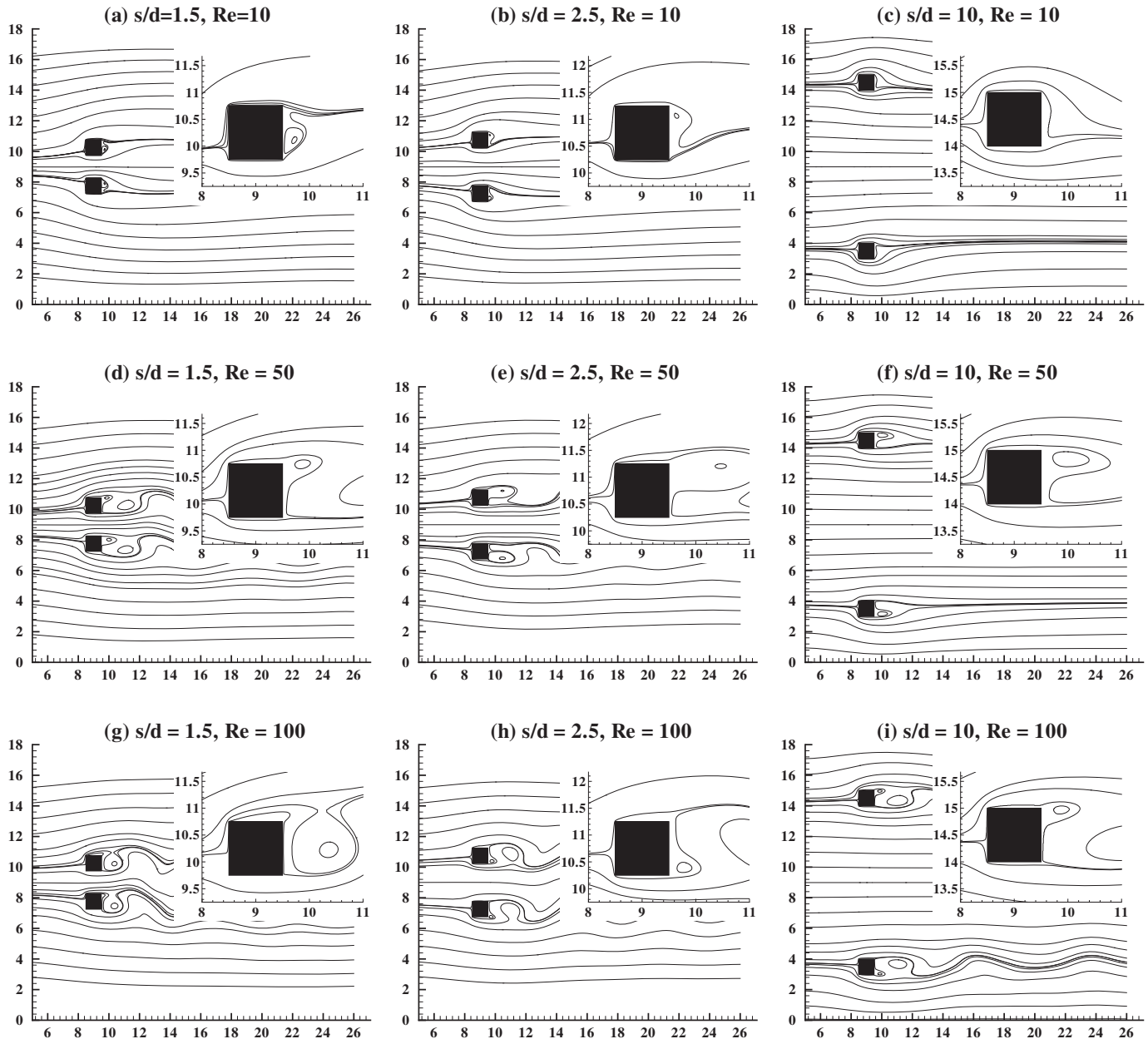
Figures 2a–2i show the flow patterns around the side-by-side square cylinders by streamline contours at different gap ratios (1.5, 2.5, 10) for various values of Reynolds numbers (10, 50, 100). The streamline contours are also shown in the magnified view for a top square cylinder in these figures. In this study, the flow is found to be steady for the range  $Re < 50$  ( $s/d = 1.5$ –2.5), for  $Re < 60$  at  $s/d = 5$  and at  $s/d = 10$ , the flow is steady for the range  $Re < 90$ . While the flow is observed to be periodic unsteady for  $Re \geq 50$  ( $s/d = 1.5$ –2.5), and for  $Re \geq 60$  at  $s/d = 5$  and at  $s/d = 10$ , it is time-periodic for the range  $Re \geq 90$ . For this analysis, the variation of lift coefficient with time is monitored for the flow to reach the fully periodic regime such that the simulations are continued until either the lift coefficient becomes zero, indicating the steady flow, or the flow becomes fully periodic in time (i.e., unsteady regime). For instance, Figures 3a–3y illustrate the instantaneous streamlines for four successive moment of time (i.e.,  $t = T_p, T_p/4, 2T_p/4$ , and  $3T_p/4$ ), which span the whole time period for  $Re = 100$ ,  $s/d = 2.5$ , and  $Pr = 0.7$ . It is observed that from the next moment onward (i.e., after  $t = 3T_p/4$ ), the flow behavior gets repeated.

The vortices are observed to disappear in the far downstream field. Also, it is worthwhile to mention that the in-phase/anti-phase vortex shedding observed here continues for all time.

It can be seen in Figures 2a–2c that the flow separates at the trailing edges of both cylinders at  $Re = 10$  for  $s/d = 1.5$ –10. As a result, a closed steady recirculating region consisting of symmetric vortices, which are in anti-phase form at the trailing edges of both the cylinders, is formed. Similar to a single square cylinder [15, 16, 22–24], the wake region increases in size with the increase in Reynolds number and the flow remains steady up to  $Re = 40$  for all  $s/d$  values used in this study. However, for  $Re = 50$  (Figures 2d and 2e), the vortices in the separation bubble start to separate alternately from the trailing edges of the square cylinders and move downstream because of Benard–von Karman instability in the unsteady periodic flow phenomenon referred to as vortex shedding for the gap ratios of 1.5 and 2.5. A similar phenomenon is also observed for the gap ratio of 2 at  $Re = 50$ , while the flow is found to be steady for the gap ratio of 5 at  $Re = 50$ . As the Reynolds number is increased from 50 to 100, the flow exhibits a synchronized in-phase regime for  $s/d = 1.5$  for the range  $50 \leq Re \leq 100$  (Figures 2d and 2g); however, for  $s/d = 2$  and 2.5, the synchronized anti-phase regime is observed for the Reynolds number range  $50 \leq Re < 90$  (Figure 2e), for  $s/d = 5$  the synchronized anti-phase regime is observed for Reynolds number range  $60 \leq Re < 90$ , and as we go from Reynolds number 90 to 100 the flow exhibits an in-phase regime for  $s/d = 2, 2.5, 5$  (Figure 2h). For  $s/d = 10$ , as the Reynolds number is increased ( $10 \leq Re < 90$ ), a steady recirculating region consisting of symmetric vortices, which are in anti-phase form, is observed (Figure 2f). However, for the Reynolds number range  $90 \leq Re \leq 100$  (at  $s/d = 10$ ), anti-phase vortex shedding is observed because of the development of the unsteady periodic flow (Figure 2i). Thus, in the 2-D laminar regime for the side-by-side square cylinders under the range of conditions studied here, there seem to be two distinct flow patterns: first, steady flow separation at the trailing edge with a separation bubble trapped behind the cylinder; and second, unsteady flow separation at the trailing edge with vortex shedding.

### Thermal Patterns

Figures 4a–4i and 5a–5i present the representative isotherms for the constant temperature condition around the two side-by-side square cylinders for  $Re = 10, 50$ , and 100 for separation ratios ( $s/d$ ) = 1.5, 2.5, and 10 at different values of Prandtl number ( $Pr = 0.7$  and 50). The isotherms are steady and symmetric for  $Re = 10$ –40 at  $s/d = 1.5$ –2.5, for  $Re = 10$ –50 at  $s/d = 5$ , and for  $Re = 10$ –80 at  $s/d = 10$  (Figures 4a–4c, 4f, and 5a–5c), whereas isotherms become periodic unsteady for  $Re = 50$ –100 at  $s/d = 1.5$ –2.5, for  $Re = 60$ –100 at  $s/d = 5$ , and for  $Re = 90$ –100 at  $s/d = 10$  (Figures 4d, 4e, 4g–4i, and 5d–5i). Figures 3e–3h illustrate the instantaneous isotherm contours for four successive moments of time for the time-periodic regime, moments that span the whole time period. Similar to the instantaneous flow

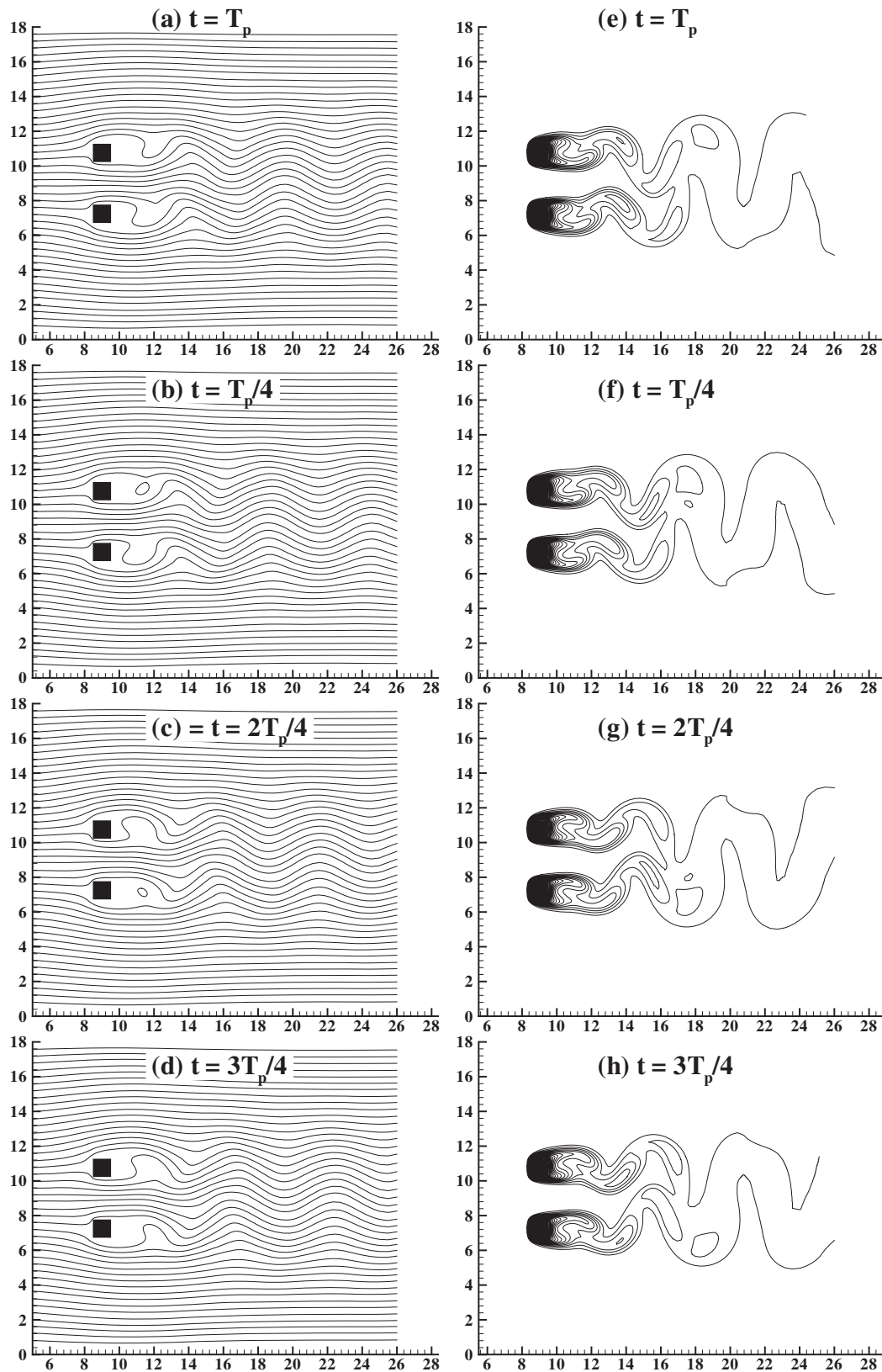


**Figure 2** Instantaneous streamline contours around the side-by-side square cylinders along with magnified views for top cylinder at different values of  $Re$  and  $s/d$ .

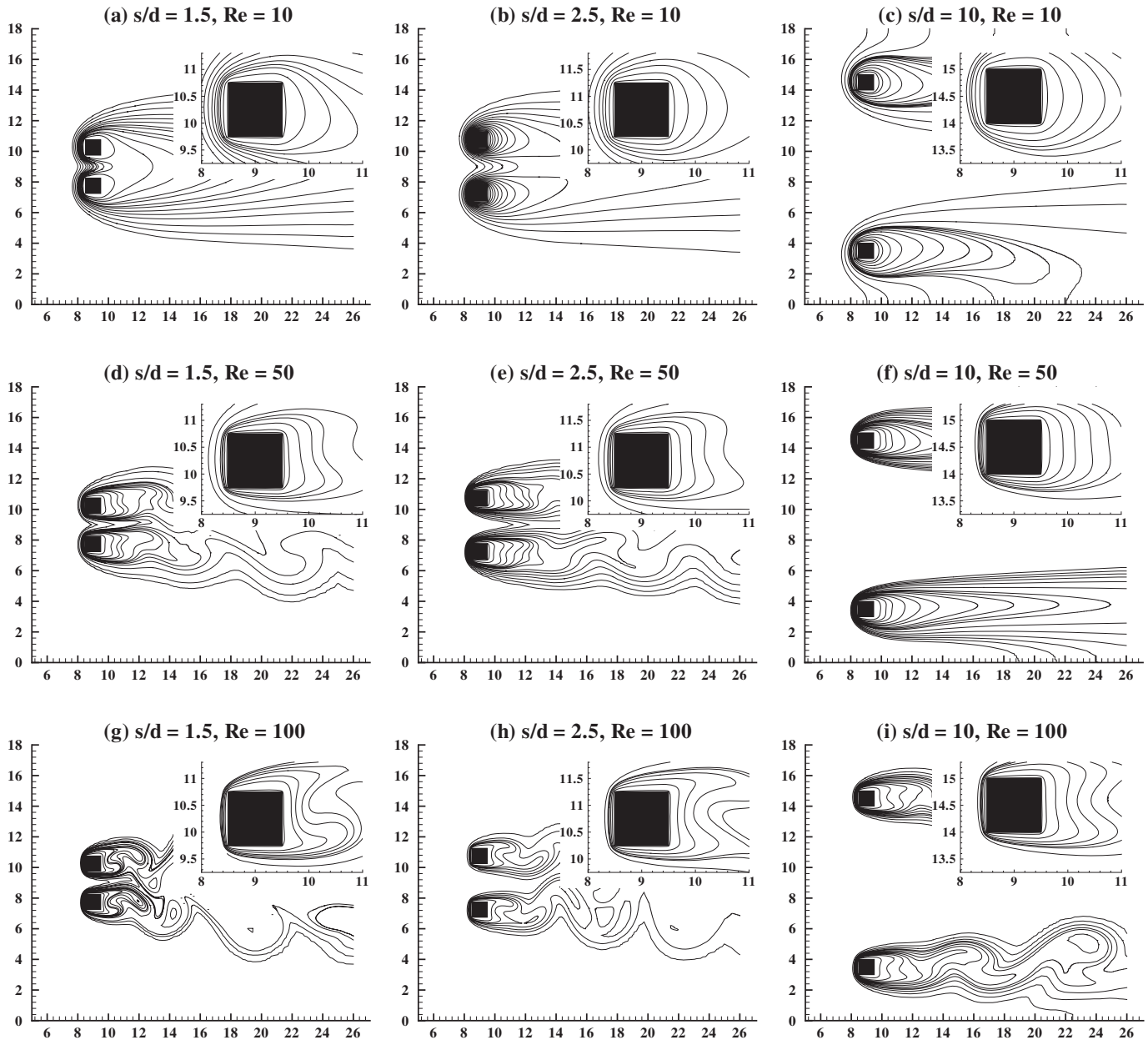
patterns, the isotherms also seem to possess the nature of alternating reversals and a temperature street behind the cylinders is formed.

Analogous to a single square cylinder [22–25], isotherms are observed to be more crowded on the front surfaces of the square cylinders, indicating higher Nusselt number values, and consequently have higher heat transfer rates on the front surfaces as compared to the other surfaces of the cylinders. Also, the clustering of isotherms near the rear surfaces of the cylinders increases with increasing Reynolds number and/or Prandtl number. This is because of the circulation of larger amount of fluid at relatively higher Reynolds number as shown in Figures 4a–4i and

5a–5i and the reduction of the thermal boundary-layer thickness for higher  $Pr = 50$  as compared to  $Pr = 0.7$  (Figure 5). Also, the temperature field is seen to decay somewhat faster for higher Prandtl number, i.e.,  $Pr = 50$  in comparison to  $Pr = 0.7$ . The isotherm contours vary significantly with Prandtl numbers as a result of a wide variation of thermal diffusivity. For the low value of the Prandtl number ( $Pr = 0.7$ ), due to enhanced thermal diffusion, the isotherms are wider around the cylinders than that of the high value of the Prandtl number. As the Prandtl number increases for a particular Reynolds number, the thermal diffusion is considerably reduced, resulting in narrow isotherm profiles around the cylinders. However, with increase in the gap



**Figure 3** Instantaneous streamline and isotherm contours around the side-by-side square cylinders for top cylinder for four successive moment of time of a periodic cycle for  $Re = 100$ ,  $s/d = 2.5$ , and  $Pr = 0.7$ .



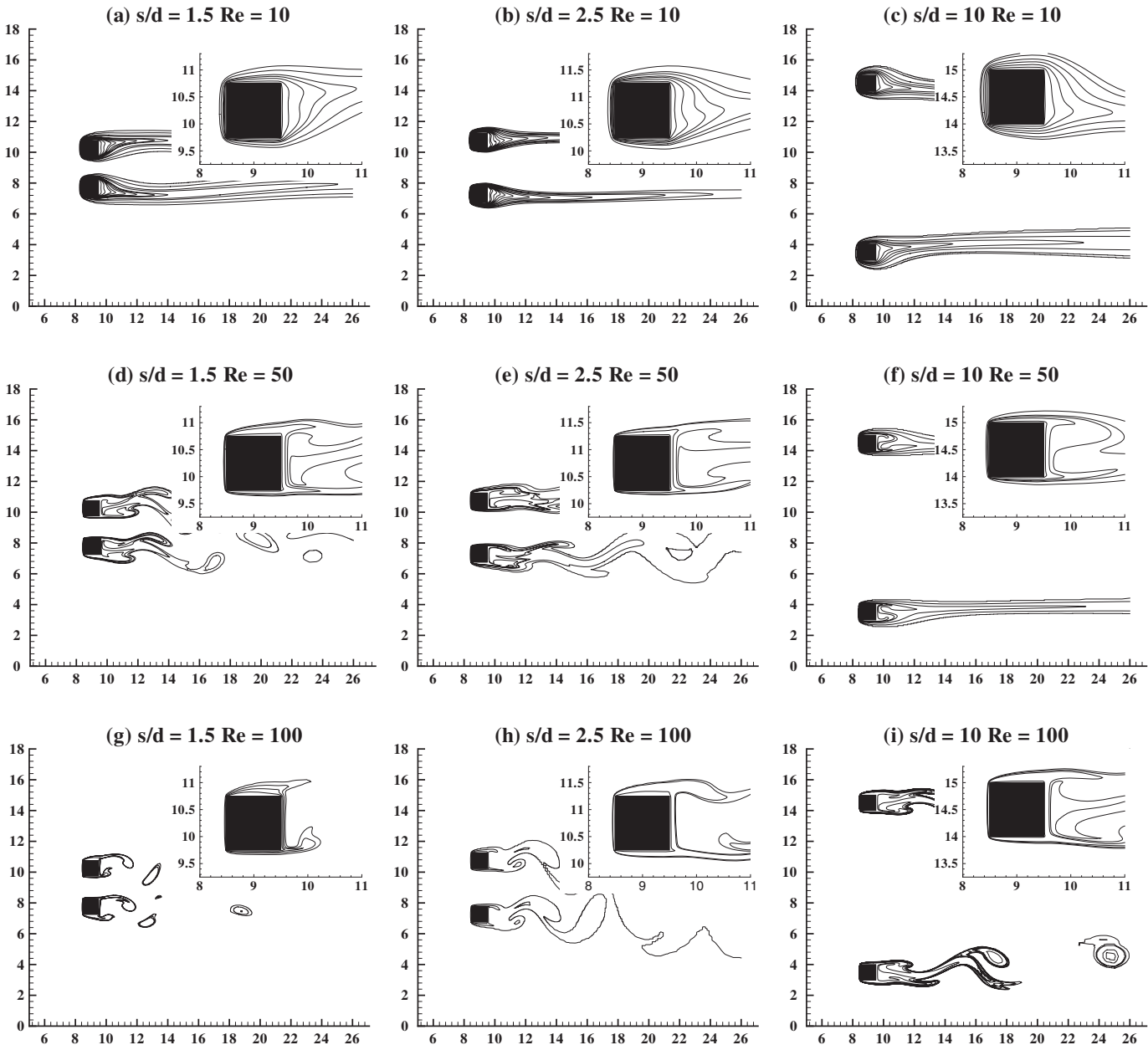
**Figure 4** Instantaneous isotherm contours around the side-by-side square cylinders along with magnified views for top cylinder for  $Pr = 0.7$  at different values of  $Re$  and  $s/d$ .

ratio, the combined isotherms for the two cylinders are separated as the gap is increased, as shown in Figures 4c, 4f, and 4i.

#### Individual and Overall Drag Coefficients

The drag is the force exerted by the fluid on the bluff body in the direction of flow. Drag force is produced by viscous friction along the surface and by an unsymmetrical pressure distribution on upstream and downstream sides of the cylinders. There are two components contributing to drag: the friction drag coefficient ( $C_{DF}$ ) and the pressure drag coefficient ( $C_{DP}$ ).

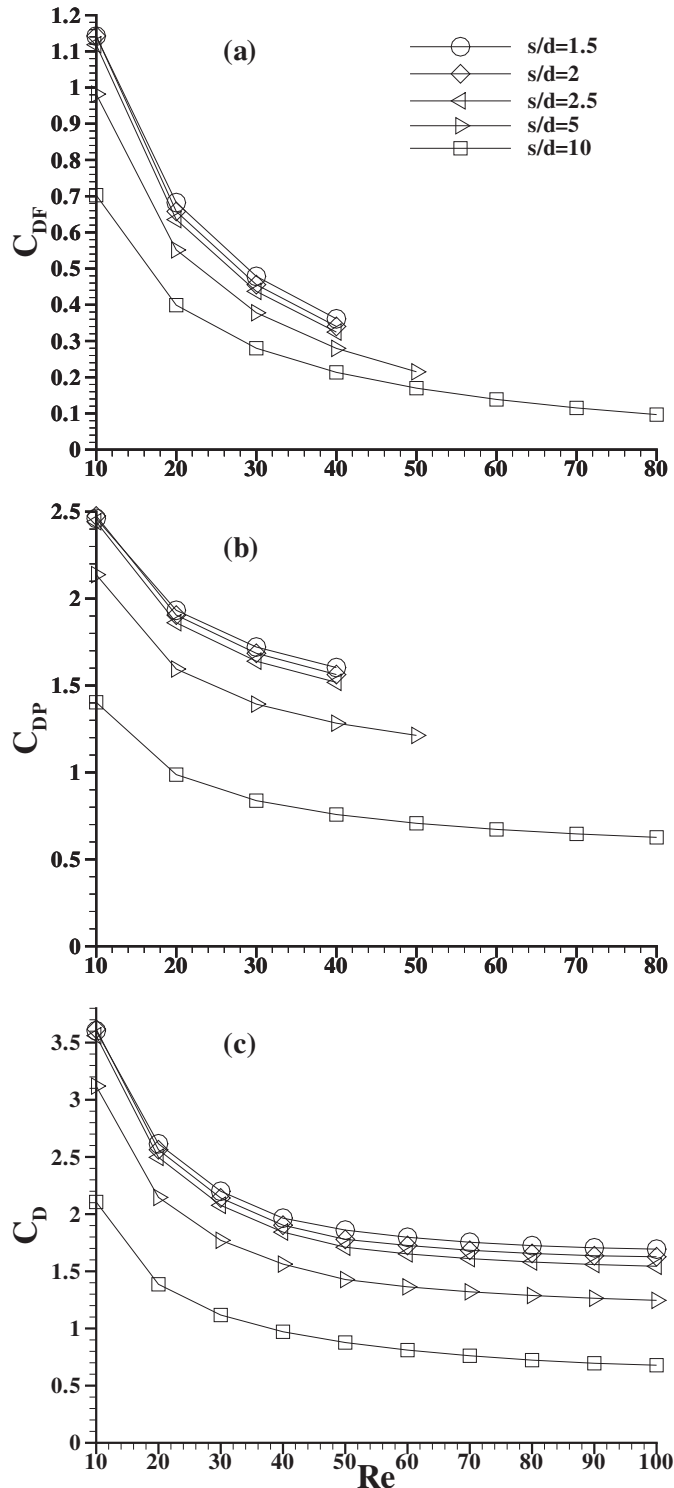
Therefore, the total drag coefficient can be written as  $C_D = C_{DF} + C_{DP}$ . It is worthwhile to mention here that identical values of drags are found for top and bottom square cylinders for the range of conditions studied. Obviously, the overall drag can be obtained by doubling the value of a top or a bottom cylinder. Figures 6a and 6b present the variation of individual drag coefficients for a top or a bottom cylinder with Reynolds numbers at various values of separation ratios in the steady regime. However, Figure 6c depicts the results for the total drag for a top or a bottom cylinder for the range of Reynolds numbers and gap ratios studied in both steady and time-periodic regimes. In this study, individual drags ( $C_{DF}$  and  $C_{DP}$ ) are calculated



**Figure 5** Isotherm contours around the side-by-side square cylinders along with magnified views for top cylinder for  $Pr = 50$  at different values of  $Re$  and  $s/d$ .

in the steady regime. However, in the time-periodic regime, individual drags are not calculated and only total drag ( $C_D$ ) is monitored with time to calculate the time-averaged value of the total drag. It is observed that the value of the drag coefficient decreases with increase in the value of the Reynolds number. Evidently, at low values of Reynolds numbers, drag is strongly dependent on the Reynolds number because the viscous forces play a more dominant role in the steady regime (i.e.,  $Re = 10$ – $40$  for  $s/d = 1.5$ – $2.5$ ,  $Re = 10$ – $50$  for  $s/d = 5$ , and  $Re = 10$ – $80$  for  $s/d = 10$ ) than in the (periodic) unsteady regime (i.e.,  $50 \leq Re \leq 100$  for  $s/d = 1.5$ – $2.5$ ,  $60 \leq Re \leq 100$  for  $s/d = 5$ ,

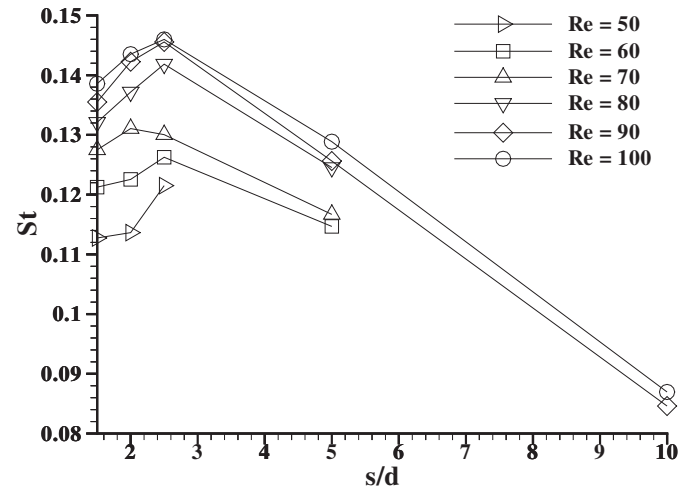
and  $90 \leq Re \leq 100$  for  $s/d = 10$ ). Also, the time-averaged drag coefficient is found to be identical for top and bottom cylinders in the time-periodic regime. Because of the symmetry of the mean flow, the values of top and bottom cylinders should be the same on average [3]. However, the instantaneous values of individual drag coefficients of top and bottom cylinders are different. Furthermore, the variation of the ratio of pressure to friction components of drag (i.e.,  $C_{DP}/C_{DF}$ ) at different values of  $Re$  and  $s/d$  reveals that the contribution of pressure drag coefficient ( $C_{DP}$ ) to total drag coefficient ( $C_D$ ) always increases with increasing Reynolds number.



**Figure 6** Variation of (a) friction and (b) pressure drag coefficients in the steady regime and (c) total drag coefficient in both steady and unsteady regimes with Re and  $s/d$ .

### Strouhal Number

The Strouhal number ( $St$ ) represents the frequency of vortex shedding in the unsteady periodic regime. The variation of the

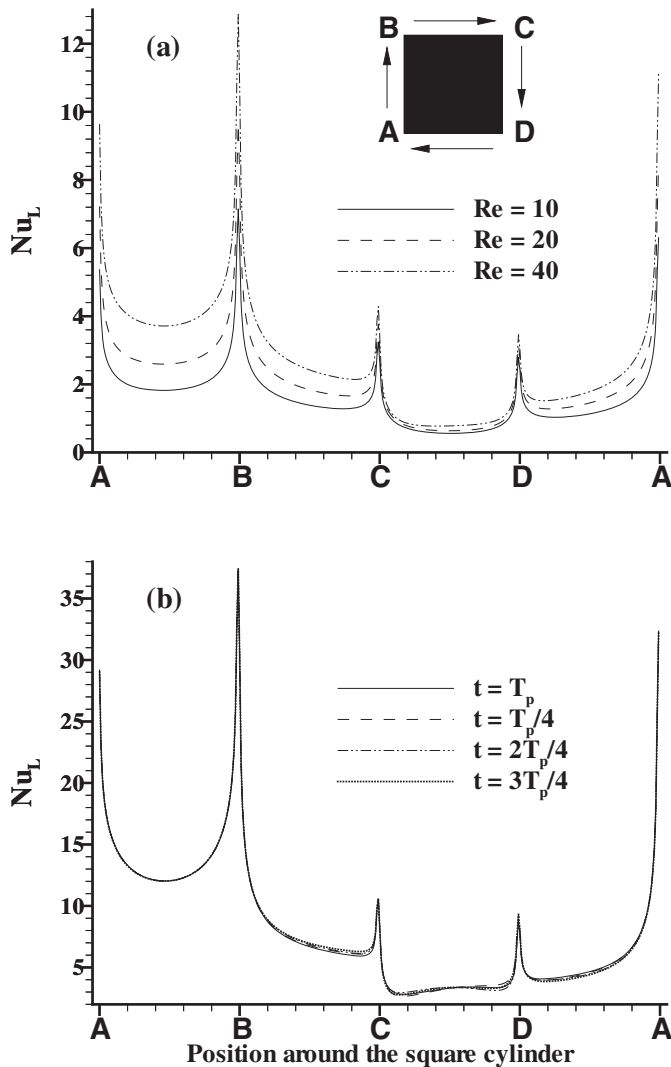


**Figure 7** Variation of Strouhal number with Reynolds number and gap ratio.

Strouhal number with respect to gap ratio and Reynolds number is presented for a top or a bottom square cylinder in Figure 7. The value of the Strouhal number is obviously zero in the steady regime (i.e.,  $10 \leq Re < 50$  for  $s/d = 1.5-2.5$ ,  $10 \leq Re < 60$  for  $s/d = 5$ , and  $10 \leq Re < 90$  for  $s/d = 10$ ), whereas in the unsteady regime ( $50 \leq Re \leq 100$  for  $s/d = 1.5-2.5$ ,  $60 \leq Re \leq 100$  for  $s/d = 5$ , and  $Re = 90-100$  for  $s/d = 10$ ), the Strouhal number increases with increase in Reynolds number for a fixed value of the gap ratio. It is also observed that for a fixed value of the Reynolds number, the value of the Strouhal number first increases and attains a maximum and then decreases with increasing value of the gap ratio. This is due to the fact that as the value of the gap ratio increases, interaction of vortices downstream of the square bars decreases. Also, beyond the gap ratio of 2.5, the two square obstacles behave as a single bluff body independently. However, the value of the Strouhal number does not go to a single cylinder value even in the limit of very large spacing of 10 used here in the side-by-side arrangement of two long square bars in a channel.

### Local and Average Nusselt Numbers

In this study, local Nusselt number ( $Nu_L$ ) on the surfaces of the long square bars is defined as  $hd/k$  and it is equal to  $-\partial\theta/\partial n_s$ . The variation of the local Nusselt number around the four surfaces of the top square cylinder is illustrated here in both steady (Figure 8a) and time-periodic (Figure 8b) regimes for  $s/d = 2.5$  and  $Pr = 0.7$ . Figure 8b shows the instantaneous variation of the local Nusselt number for the four successive moments of time that span the entire time period. The variation of the local Nusselt around a top or a bottom square cylinder is observed to be similar to a single square cylinder [22–25]. The local Nusselt number is observed to be the highest for the front surface where the incoming fluid is in direct contact with this surface, the lowest for the rear surface, and intermediate for the

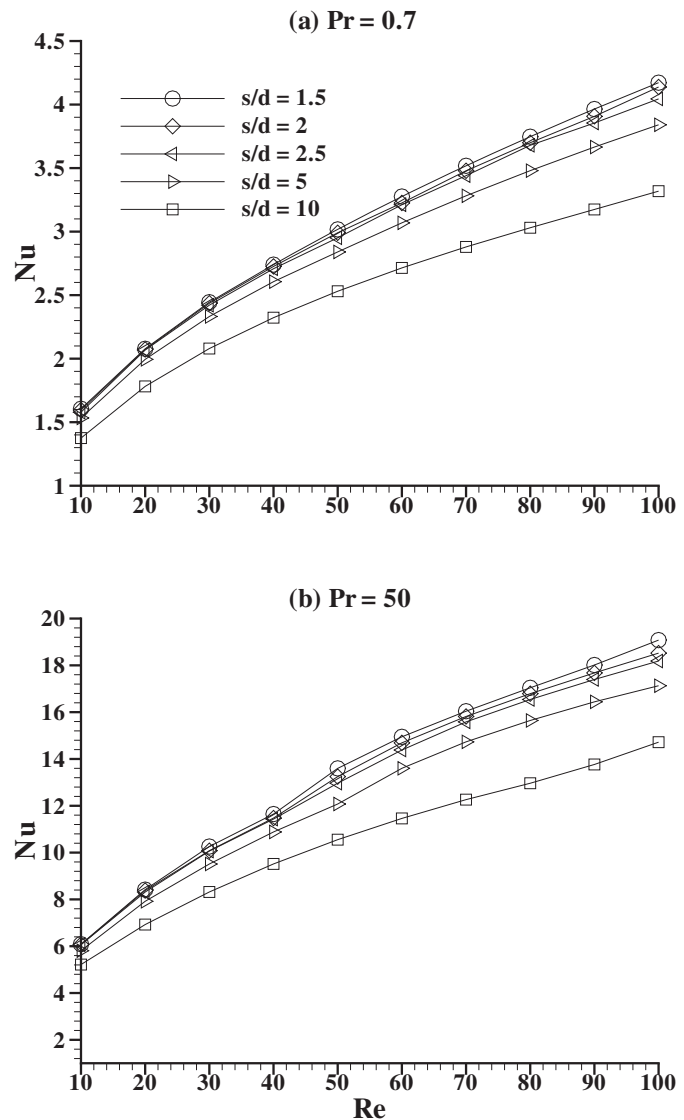


**Figure 8** Variation of local Nusselt number for (a)  $Re = 10, 20$ , and  $40$  in the steady regime and for (b)  $Re = 100$  in the time-periodic regime for  $s/d = 2.5$  and  $Pr = 0.7$ .

side surfaces. As expected, the local Nusselt number increases with increasing Reynolds number.

The average Nusselt number is obtained by averaging the local Nusselt number on the surfaces of cylinders. The variation of the average Nusselt number for a top or a bottom cylinder with Reynolds number for the values of Reynolds numbers ( $Re$ ) = 10–100 and Prandtl numbers ( $Pr$ ) = 0.7–50, for all the separation ratios, is presented in Figures 9a and 9b. It is seen from these figures that the average Nusselt number for the square cylinders increases with increasing Reynolds number and/or Prandtl number for all the separation ratios studied. With an increase in the value of the separation ratio, the average Nusselt number of both square cylinders decreases.

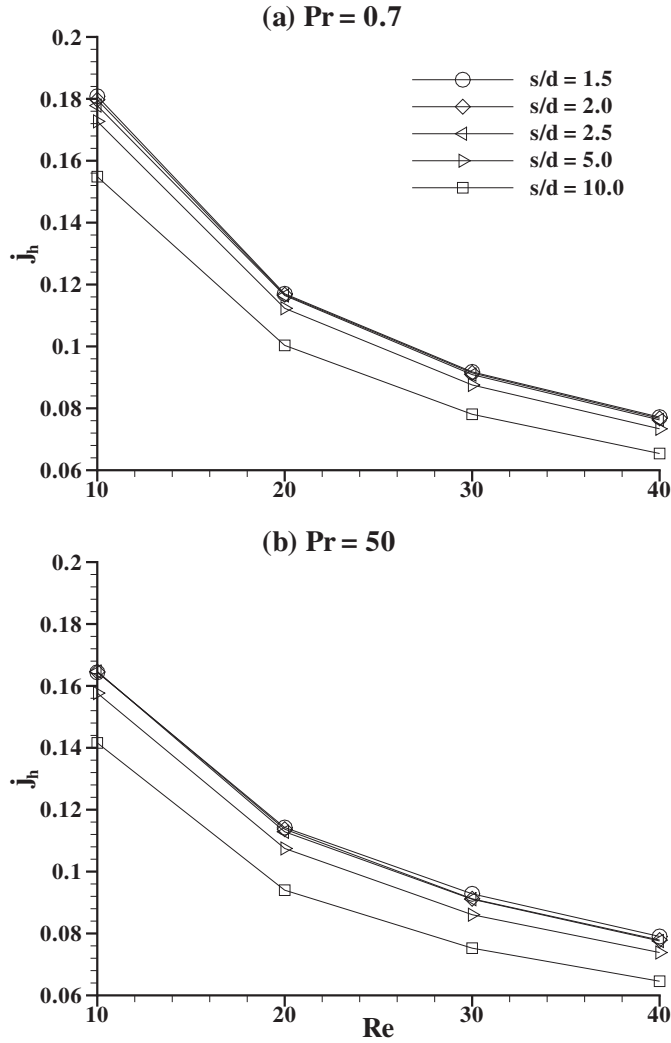
As the heat transfer enhancement from any bluff body is associated with the downstream vortex structures, the percentage maximum enhancement in the value of the average Nusselt number is calculated as  $(|Nu_{Pr=50} - Nu_{Pr=0.7}|)/Nu_{Pr=50} \times 100$  and is



**Figure 9** Variation of average Nusselt number for (a)  $Pr = 0.7$  and (b)  $Pr = 50$  at different values of Reynolds number and gap ratio.

found to be approximately 76.56% for the range of conditions investigated. The enhancement in heat transfer is because of the generation of vortices due to the long square bars and the channel walls. The enhancement in the value of the average Nusselt number decreases slightly with the increasing value of the gap ratio.

Furthermore, from an application point of view, it is convenient to correlate the present heat transfer results by using Colburn heat transfer factor ( $j_h$ ). Simple dimensional considerations suggest that the Nusselt number is a function of Reynolds number, Prandtl number, and gap ratio. The  $j_h$  factor is defined as  $j_h = Nu/(Re \times Pr^{1/3})$ . The variation of  $j_h$  factor with Reynolds number, Prandtl number, and gap ratio is shown in Figures 10a and 10b. The heat transfer  $j_h$  factor collapses the results for different values of Prandtl numbers onto one curve and it can also be used to predict the heat transfer rates at different Reynolds



**Figure 10** Variation of Colburn  $j_h$  factor for (a)  $Pr = 0.7$  and (b)  $Pr = 50$  at different values of Reynolds number and gap ratio in the steady regime.

numbers. Clearly, as the Reynolds number increases, the Colburn heat transfer factor decreases in the steady regime. Further, the following simple expression (Eq. 5) correlates the present heat transfer results adequately for the constant-temperature case in the steady flow regime for the range  $Re = 10\text{--}40$ ,  $s/d = 1.5\text{--}5$ , and  $Pr = 0.7\text{--}50$ . However, a separate correlation (Eq. 6) is provided at the gap ratio ( $s/d = 10$ ) for the described ranges of  $Re$  and  $Pr$  as the significant change in the heat transfer values for this gap ratio is observed.

For  $s/d = 1.5\text{--}5$ ,

$$j_h = 0.631 \times Re^{-0.575} \quad (5)$$

For  $s/d = 10$ ,

$$j_h = 0.585 \times Re^{-0.595} \quad (6)$$

The maximum percentage deviations of Eqs. (5) and (6) with present computed values are found to be less than 5.0%. However, the maximum deviations are about 7.0% for  $1.5 < s/d \leq 5$  at  $Re = 10$ .

## CONCLUSIONS

In the present study, a 2-D incompressible numerical investigation is conducted to analyze the laminar flow and heat transfer characteristics in a confined channel with two square cylinders (of equal diameters) arranged side-by-side. The analysis is carried out for Reynolds numbers ( $Re = 10\text{--}100$ ) (thereby covering both steady and time-periodic regimes) and Prandtl numbers ( $Pr = 0.7\text{--}50$ ) for transverse separation distance between the long square bars ( $s/d = 1.5, 2, 2.5, 5$ , and  $10$ ). The flow is found to be steady for the range  $Re < 50$  ( $s/d = 1.5\text{--}2.5$ ), and for  $Re < 60$  at  $s/d = 5$  and at  $s/d = 10$ , and it is steady for the range  $Re < 90$  at  $s/d = 10$ . While the flow is (periodic) unsteady for  $Re \geq 50$  ( $s/d = 1.5\text{--}2.5$ ), and for  $Re \geq 60$  at  $s/d = 5$  and at  $s/d = 10$ , it is unsteady for  $Re \geq 90$ . The streamline and isotherm contours are represented to understand the flow and thermal patterns around the long square obstacles, respectively. The maximum crowding of isotherms is found to be on the front surface, indicating the highest Nusselt number, as compared to other surfaces of the cylinders. The overall drag coefficient decreases with increasing Reynolds number, whereas the average Nusselt number increases with increasing Reynolds number and/or Prandtl number for all the separation ratios studied here. The percentage enhancement in the value of the average Nusselt number is found to be more than 76% for the range of conditions embraced in this work.

## NOMENCLATURE

|           |                                                                              |
|-----------|------------------------------------------------------------------------------|
| $c_p$     | specific heat of the fluid, $J\ kg^{-1}\ K^{-1}$                             |
| $C_p$     | pressure coefficient                                                         |
| $C_{DF}$  | friction drag coefficient                                                    |
| $C_{DP}$  | pressure drag coefficient                                                    |
| $C_D$     | total drag coefficient ( $= \frac{2F_D}{\rho U_{\max}^2 d}$ )                |
| $C_L$     | total lift coefficient ( $= \frac{2F_L}{\rho U_{\max}^2 d}$ )                |
| $d$       | side of a square cylinder, m                                                 |
| $f$       | vortex shedding frequency, $s^{-1}$                                          |
| $F_D$     | total drag force per unit length of the cylinder, $N\ m^{-1}$                |
| $F_{DF}$  | frictional component of the total drag force, $N\ m^{-1}$                    |
| $F_{DP}$  | pressure component of the total drag force, $N\ m^{-1}$                      |
| $F_L$     | total lift force per unit length of the cylinder, $N\ m^{-1}$                |
| $h$       | local heat transfer coefficient, $W\ m^{-2}\ K^{-1}$                         |
| $\bar{h}$ | average heat transfer coefficient, $W\ m^{-2}\ K^{-1}$                       |
| $H$       | height of the computational domain, m                                        |
| $j_h$     | Colburn heat transfer factor ( $= \frac{Nu}{Re\ Pr^{1/3}}$ )                 |
| $k$       | thermal conductivity of the fluid, $W\ m^{-1}\ K^{-1}$                       |
| $L$       | length of the computational domain, m                                        |
| $n_s$     | unit vector normal to the surface of the cylinder                            |
| $n_x$     | $x$ -component of the direction vector normal to the surface of the cylinder |
| $Nu_L$    | local Nusselt number ( $= \frac{hd}{k}$ )                                    |
| $Nu$      | average Nusselt number ( $= \frac{\bar{h}d}{k}$ )                            |
| $p$       | pressure, $N\ m^{-2}$                                                        |

|              |                                                                         |
|--------------|-------------------------------------------------------------------------|
| Pr           | Prandtl number ( $= \frac{\mu c_p}{k}$ )                                |
| Re           | Reynolds number ( $= \frac{d U_{\max} \rho}{\mu}$ )                     |
| s            | transverse separation distance between the square bars, m               |
| S            | surface area, m <sup>2</sup>                                            |
| St           | Strouhal number ( $= \frac{f d}{U_{\max}}$ )                            |
| t            | time, s                                                                 |
| T            | temperature, K                                                          |
| $T_p$        | time period for one cycle, s                                            |
| $T_w$        | constant wall temperature at the surface of the cylinder, K             |
| $T_{\infty}$ | temperature of the fluid at the inlet, K                                |
| $U_{\max}$   | maximum velocity at the channel inlet, m s <sup>-1</sup>                |
| $V_x$        | component of velocity in the x-direction ( $= \frac{V_x^*}{U_{\max}}$ ) |
| $V_y$        | component of velocity in the y-direction ( $= \frac{V_y^*}{U_{\max}}$ ) |
| x            | stream-wise coordinate ( $= \frac{x^*}{d}$ )                            |
| $X_d$        | downstream distance of the cylinder, m                                  |
| $X_u$        | upstream distance of the cylinder, m                                    |
| y            | transverse coordinate ( $= \frac{y^*}{d}$ )                             |

### Greek Symbols

|          |                                                             |
|----------|-------------------------------------------------------------|
| $\beta$  | blockage ratio ( $= \frac{d}{H}$ )                          |
| $\theta$ | temperature ( $= \frac{T - T_{\infty}}{T_w - T_{\infty}}$ ) |
| $\rho$   | density of the fluid, kg m <sup>-3</sup>                    |
| $\mu$    | viscosity of the fluid, kg m <sup>-1</sup> s <sup>-1</sup>  |
| $\delta$ | the smallest grid size, m                                   |
| $\tau$   | shear stress acting on the surface dS, Pa                   |

### Superscript

\* dimensional variable

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